

Mohit Dubey

AI Full Stack Engineer

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SUMMARY

Senior Full Stack Developer with 5+ years of experience designing and building high-performance **React** applications and **Python/Node.js** backend services, paired with deep expertise in **AI agent** and **LLM application** development end-to-end — including agentic orchestration logic (**LangChain**, **LangGraph**, **ReAct**-style reasoning/action loops). Skilled in **REST/GraphQL API integration**, agent tooling, and **observability/tracing** (**Arize**, **Datadog**) for production agent systems, with **CI/CD** automation via **GitHub Actions** and **Jenkins**. Comfortable owning architecture end-to-end in **Agile/Scrum** teams, with hands-on experience on the **Databricks** platform.

WORK EXPERIENCE

Software Engineer — Publicis Sapient

June 2025 - Present

- Led the architecture and development of an enterprise-grade **GraphRAG**-based private coding assistant for a B2B SaaS platform using **LangGraph**, **DeepAgents**, and agentic AI, improving code retrieval accuracy by 38% and reducing developer resolution time by 60%.
- Designed and implemented an **MCP server** with dynamic tool discovery, semantic search, and context orchestration, reducing LLM context overhead by 70% and lowering token consumption and inference costs across production workloads.
- Integrated **LiteLLM** as a unified LLM gateway to standardize routing across multiple model providers, enabling cost-based fallback, rate limiting, and centralized usage tracking across agent workloads.
- Instrumented production LLM agents with **Arize** for real-time evaluation, drift detection, and trace-level observability, and **Datadog** for infrastructure/APM monitoring, cutting mean-time-to-detect on model regressions.
- Automated build, test, and deployment pipelines using **GitHub Actions** and **Jenkins**, enabling zero-downtime releases and consistent rollout of agentic services across environments.
- Built a **multi-tenant, multi-agent AI platform** with real-time guardrails, memory management, and policy enforcement, enabling secure deployment across **multiple tenants** while maintaining sub-second response latency.
- Architected scalable agent workflows leveraging **LangGraph**, vector databases, reranking, and hybrid retrieval techniques, increasing answer relevance by 42% and supporting thousands of concurrent enterprise requests.
- Designed and implemented a **client-server UI version synchronization system** (React + Nginx) to ensure users always run the latest deployed build.
- Built a **reusable, highly configurable table component** adopted across 15+ modules, improving consistency and reducing maintenance overhead.
- Centralized API interactions by creating a **shared network layer** with consistent error handling and retry logic, improving reliability across the application.
- Implemented **performance-optimization** techniques such as **code-splitting**, **lazy loading**, and **dynamic imports** to reduce **JavaScript bundle size** and improve time to interactive (TTI).
- Engineered scalable **Frontend Architecture** using **React.js**, **Next.js**, and micro-frontends, optimizing platform performance by **60%** and securing workflows using **OAuth2** authentication protocols.

Software Engineer — Deqode

Aug 2023 - May 2025

- Architected enterprise-grade **RAG pipeline** using **LangChain**, Pinecone/FAISS, and AWS, reducing LLM hallucinations by 32% and improving retrieval latency by 45% across 1M+ documents.
- Fine-tuned 7B/13B **Qwen3-VL** LLMs using **SFT** and **QLoRA** on **AWS SageMaker** GPU clusters, improving domain-specific accuracy by 21% while cutting inference cost by 60%.
- Deployed production **LLM serving infrastructure** on **GCP Vertex AI** with **vLLM** and TensorRT-LLM, serving 13B parameter models at sub-200ms latency with 40% cost reduction for 500K+ requests/day.
- Orchestrated cloud-native ML systems on **Kubernetes** with **MLflow**, Terraform, and Vertex AI Pipelines; led platform modernization for **15+ ML teams**, compressing model deployment cycle from 3 weeks to 2 days.
- Architected **distributed training** using DeepSpeed and FSDP for multi-GPU foundation model runs, reducing training time by 65% and scaling to 34B+ parameter models.
- Deployed fluid, multi-agent conversational systems leveraging LangChain Agents, Streamlit UIs, and AWS Bedrock API gateways; **mentored AI engineering teams** on RAG scaling laws and context-window optimization.
- Created robust **AI monitoring and real-time dashboarding infrastructures** using Grafana, Power BI, and Looker to continuously track operational metrics, system health, and model drift.
- Translated deep AI behavioral insights into predictive analytics models, enhancing customer experience metrics and optimizing down-stream system performance.
- Built Aadhaar-Voter ID linking UI in React, scaling to **scaling to handle high-traffic production loads** with resilient, reusable components.
- Migrated **Redux** to modular **RTK slices** + RTK Query, reducing API overfetching by 78%.
- Developed editable virtualized grid (1M+ rows) using **react-window**, with smooth scroll and memoization.
- Led federated design **system adopted by 50+ teams**, enforced standards.
- Integrated Web Vitals + **Sentry** to track LCP, FCP, TTI across 500k+ sessions/month and optimize core web metrics.
- Built **secure file uploader** with presigned S3, retry/resume, chunked upload, drag-and-drop UI, and Fetch API.

- Implemented **keyboard navigation**, focus management, ARIA roles, and screen reader support using semantic HTML.
- Built schema-to-React renderer supporting **dynamic layouts**, conditional fields, async data bindings, and versioned JSON configs; cut UI rollout time.

Software Engineer — Shiv Technolabs

Feb 2023 - July 2023

- Leveraged **deep knowledge of browser internals** to optimize performance, systematically reducing reflow,repaint, and bundle sizes to achieve significant improvements in Core Web Vitals.
- Implemented **advanced state management** patterns across complex application workflows using MobX and modern ES6+ JavaScript features for predictable data flow.
- Led **frontend performance audits** analyzing JS engines compilation behaviors and execution contexts to eliminate memory leaks and runtime bottlenecks.
- Orchestrated the transition to **Next.js** methodologies, integrating **RESTful APIs** to achieve seamless data transfer across platforms, which decreased website bounce rate by **15%** and grew traffic.

Software Engineer — Sourcebae

May 2021 - Jan 2023

- Built full-stack applications with **React**, **TypeScript**, and **NestJS**, shipping REST API backends backed by **PostgreSQL** and **MongoDB** serving 100K+ users with sub-100ms p99 latency via Redis caching.
- Designed a reusable **TypeScript** component library with centralized state management, cutting feature delivery time by 40% across 5 concurrent product teams.
- Containerized services with **Docker** and established **CI/CD pipelines** with automated testing and staged rollouts, eliminating manual deployment errors and enabling zero-downtime releases.
- Built **offline-first UI** in React Native using SQLite for local data storage and background sync to update the server when reconnected.

PROJECTS

TradeBrains — Automated Real-Time Algorithmic Trading Platform

- Architected an end-to-end algorithmic trading platform predicting 20-minute stock movements by fine-tuning open-source generative models (**Llama 3.3 70B**) using 3-stage **RLHF (SFT, Reward Model, PPO)** and **GRPO** to align prediction outputs directly with target performance metrics.
- Implemented custom **GRPO/PPO reward functions** integrating multi-objective metrics (factual correctness, formatting compliance, and financial risk metrics) to eliminate hallucinations in high-stakes domain outputs.
- Engineered **QLoRA 4-bit** parameter-efficient fine-tuning (**PEFT**) pipelines for 8B-parameter sentiment models, reducing memory footprints while maintaining **88% structured score extraction accuracy**.
- Executed **Direct Preference Optimization (DPO)** and **KTO** fine-tuning on domain-specific instruction datasets, bypassing separate reward model latency while matching RLHF alignment quality.

EDUCATION

Rajiv Gandhi Proudhyogiki Vishwavidyalaya

Bhopal, India

Bachelor of Engineering (HONOURS) - Information Technology; GPA: 7.75

August 2016 - May 2020

ACHIEVEMENTS

- [AWS Certified Developer Associate](#).
- [AWS Certified AI Practitioner](#).
- [AWS Certified Cloud Practitioner](#).
- [Hacker Rank Javascript Basic](#).
- [Hacker Rank CSS certification](#).
- [Sololearn React + Redux](#)

TECHNICAL SKILLS

- **LLMs & Agents:** LangChain, LangGraph, CrewAI, Google ADK, LlamaIndex, Hugging Face Transformers, OpenAI API, Ollama, vLLM, TensorRT-LLM, LiteLLM, MCP Servers, Agent Orchestration, RAG Systems, Prompt Engineering
- **Training & Optimization:** LLM Fine-Tuning (SFT, QLoRA, LoRA, PEFT), RLHF, GRPO, DeepSpeed, FSDP, Distributed Training, CUDA, Model Quantization (INT8, NF4)
- **ML & Deep Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, Classical ML, Feature Engineering, Model Optimization
- **Data Engineering:** Apache Spark (PySpark, Spark SQL), Databricks, Delta Lake, Kafka, Airflow, BigQuery, ETL/ELT, Feature Store (Feast)
- **MLOps & Infrastructure:** MLflow, Kubeflow, Kubernetes, Docker, Terraform, CI/CD Pipelines (GitHub Actions, Jenkins), Model Monitoring & LLM Observability (Arize, Datadog), Drift Detection, A/B Testing
- **Cloud:** AWS (SageMaker, EKS, Lambda, S3), GCP (Vertex AI, GKE, Dataproc, BigQuery)
- **Backend Engineering:** Python, FastAPI, Go, TypeScript, REST APIs, gRPC, Microservices, WebSocket, Redis, n8n
- **Databases & Vector Stores:** PostgreSQL, MongoDB, Redis, Qdrant, Milvus, Weaviate, Chroma, SQL
- **Frameworks:** Reactjs, Redux, Nextjs, SCSS, MaterialUI, ExpressJs, Jest, RTL, Enzyme
- **Tools:** Azure, Docker, GitHub Actions, Jenkins, Datadog, Webpack, Git/Github, Postman, Swagger, Figma, Sequelize ORM, Firebase

- **Coursework:** System Design, Database Management Systems, Data Structures and Algorithms, Object Oriented Programming
- **Other:** RESTful Web Services, Agile methodologies, Integration testing, micro frontends